

■ 3.3.5 Polynomials Modulo Primes

Mathematica can work with polynomials whose coefficients are in the finite field Z_p of integers modulo a prime p .

<code>Mod[poly, n]</code>	reduce the coefficients in a polynomial modulo n
<code>Factor[poly, Modulus->p]</code>	factor $poly$ modulo p
<code>GCD[poly₁, poly₂, ..., Modulus->p]</code>	find the gcd of the $poly_i$ modulo p

Functions for manipulating polynomials over finite fields.

Here is an ordinary polynomial.

```
In[1]:= Expand[ (1 + x)^6 ]
Out[1]= 1 + 6 x + 15 x^2 + 20 x^3 + 15 x^4 + 6 x^5 + x^6
```

This reduces the coefficients modulo two.

```
In[2]:= Mod[%, 2]
Out[2]= 1 + x^2 + x^4 + x^6
```

Here are the factors of the resulting polynomial over the integers.

```
In[3]:= Factor[%]
Out[3]= (1 + x^2) (1 + x^4)
```

If you work modulo two, further factoring becomes possible.

```
In[4]:= Factor[%, Modulus->2]
Out[4]= (1 + x)^6
```